
High Frequency On-Off Grid Inverter Communication protocol V1.23

Revise Historical Records

Date	Edition	Explain	Author	Check and approve
2023/12/22	V1.2	1. The loads power is displayed as the actual value. 2. Add mains input power and generation display 3. Add the PV2 flag 4. Add the PYL-A lithium battery agreement option		
2024/06/07	V1.21	1. Add the HFP model on-grid current setting		
2024/02/26	V1.23	Add Display terminal value		

1. Overview

This protocol is used to regulate the communication between XINDUN UPS, EPS and INV devices and background PC software. The protocol refers to the implementation of MODBUS standard on TCP/IP, and has made some improvements to the communication mechanism and communication management according to the characteristics of XINDUN devices.

1.1. References

GB/T 19582-1 2008 Industrial Automation Network Specifications Based on Modbus Protocol Part 1: Modbus Application Protocol

GB/T 19582-2 2008 Industrial Automation Network Specifications Based on Modbus Protocol Part 3: Guide to the Implementation of Modbus Protocol on Serial Links

1.2. Abbreviations

UPS: Uninterruptible power supply system

EPS: Emergency power supply system

INV: Inverter

2. Communication mode

2.1. Physical layer interface

RS485 communication. Port rate 9600bps, 1 start bit, 1 stop bit, no parity.

2.2. Communication mechanism

The device adopts serial port MODBUS RTU communication mode and acts as a service (slave).

3. Information type and information structure

3.1. Data types

Table 3-1 Data type table

Data type	Symbol	Explain
An 8-bit unsigned integer	BYTE	0~255
16-bit unsigned integer	UINT16	0~65535
16-bit signed integer	INT16	-32768~32767
32-bit floating-point	FLOAT	IEEE754 short real numbers
32-bit unsigned integer	UINT32	
32-bit signed integer	INT32	
64-bit unsigned integer	UINT64	

64-bit signed integer	INT64	
A [N-byte] string	String[N]	
[N bytes] byte array	BYTEA[N]	
Boolean	BOOL	There are only two states, "0" and "1". It is used only for discrete input registers.

3.2. Command structure

Table 3-2 Command structure table

Field	Number of bytes	Description
address	1	Byte data, the address of an object. Range 0 to 247.0 is the broadcast address. 247 is the universal point-to-point communication address (only for point-to-point communication links).
FC	1	Byte data. See "command parsing". Under normal circumstances, the function code is the same for sending and answering. For abnormal circumstances, refer to "exception response mechanism"
data	N	See "Command parsing".
verification	2	MODBUS CRC16 check is adopted, and the high byte is at the front

3.3. Data storage and transmission order

3.3.1. Byte storage and transmission order of 16-bit integer type

For 16-bit integers, the high byte is transmitted first and the low byte is transmitted next. For example, 3A56H is transmitted first as 3AH and then as 56H.

Input registers and holding registers are both 16-bit addressable registers. The bit storage order for a single register can be considered to be high-order byte first, low-order byte last.

3.3.2. 32-bit integer type word storage and transmission order

For 32-bit integers, the high 16 bits are transmitted first, followed by the low 16 bits. For example, the data 8DF377A2H is transmitted first as 8DF3H, followed by 77A2H.

In input and hold registers, 32-bit data occupies two consecutive register addresses. The lower address stores the high 16 bits, while the higher address holds the low 16 bits. For example, if the value 8DF377A2H is stored using registers 12 and 13, register 12 would store 8DF3H, and register 13 would store 77A2H.

3.3.3 Floating point number storage and transmission order

Floating-point numbers use the IEEE 32-bit standard floating-point format (standard C language format), are 32 bits long, and are stored in four bytes. The format is shown in the following table.

D31	D30-D23	D22-D16	D15-D8	D7-D0
Floating-point sign S	The sequence number	High end of the tail	Median in the tail	Low tail position

If the exponent is E and the tail number is M, then the floating-point value is $\pm (1+M \times 2^{-23}) \times 2^{E-127}$

The sign of a floating point number depends on the value of the sign bit S, where S=1 indicates a negative floating point number and S=0 indicates a positive floating point number.

For example, when the 32-bit floating-point number is 40H, A0H,00H,00H, S=0, E=81H=129, M=200000H=221, then:

Floating point value = $(1+221 \times 2^{-23}) \times 2^{129-127} = 5.0$.

Four bytes are sent in the order of sign bit, exponent, high bit of decimal number, middle bit of decimal number and low bit of decimal number.

4. Command parsing

4.1. Read the hold register 03H

Host command format:

Definition	Address	FC	Register starting address	Number of registers	CRC verification
data	ADDR	03H	sREG	nREG	CRC 16
Number of bytes	1	1	2	2	2

Server response format:

Definition	Address	FC	Returns the number of data bytes	Return data	CRC verification
data	ADDR	03H	X	DATA	CRC 16
Number of bytes	1	1	1	N*2	2

4.2. Write to the single hold register 06H

Host command format:

Definition	Address	FC	Register address	Data	CRC verification
data	ADDR	06H	sREG	DATA	CRC 16
Number of bytes	1	1	2	2	2

Server response format:

Definition	Address	FC	Register address	Data	CRC verification
Number of bytes	1	1	2	2	2

4.3. Write to multiple holding registers 10H

Host command format:

Definition	Address	FC	Start address	Number of write registers	Number of data bytes	Data	CRC verification
data	ADDR	10H	sREG	nREG	X	DATA	CRC 16
Number of bytes	1	1	2	2	1	X	2

Server response format:

Definition	Address	FC	Start address	Number of write registers	CRC verification
data	ADDR	10H	sREG	nREG	CRC 16
Number of bytes	1	1	2	2	2

5. Register definitions

Register definitions

Register address	Register Name	data type	read and write	Unit	coefficient	Explanation
0	Adaptive Protocol Identification	UINT16	R			1:High frequency on-off grid inverter
0X0001	Inverter current 499	UINT16	R	A	0.01	
0X0002	inverter voltage	UINT16	R	V	1	
0X0003						
0X0004	bus voltage	UINT16	R	V	0.1	
0X0005	Input Voltage	UINT16	R	V	1	
0X0006	Load current	UINT16	R	A	0.1	
0X0007	Input current 494	UINT16	R	A	0.1	
0X0008	Battery voltage	UINT16	R	V	0.1	
0X0009	bus current	INT16	R	A	0.01	
0X000A	Charging temperature	UINT16	R	℃	1	
0X000B	PV1 current	UINT16	R	A	0.1	
0X000C	PV1 voltage 489	UINT16	R	V	0.1	
0X000D	PV2 current	UINT16	R	A	0.1	
0X000E	PV2 voltage	UINT16	R	V	0.1	
0X000F	DC temperature	UINT16	R	℃	1	
0X0010	inverter temperature	UINT16	R	℃	1	
0X0011	Battery charging and discharging current 484	INT16	R	A	0.1	
0X0012	Input frequency	UINT16	R	Hz	0.1	
0X0013	Load Percentage	UINT16	R	%	1	
0X0014	Device Working Mode	UINT16	R			0X33:Mains Power Priority Mode 0X66:PV Priority Mode 0X99:Battery Priority Mode
0X0015	Working Mode of Inverter Section	UINT16	R			0X44:Mains power supply mode 0X55:On Grid mode 0X88:Inversion Mode 0XAA:Rectification Mode 0X00:Stop working
0X0016	DCDCpartial mode 479	UINT16	R			0X55:charging mode 0XAA:discharging mode 0X00:stop working
0X0017	System flag bits	UINT16	R			
0X0018	Inverter Flag Bit	UINT16	R			
0X0019	DC flag bit	UINT16	R			
0X001A	MPPT Flag bit	UINT16	R			
0X001B	bus voltage reference	UINT16	R	V	0.1	Test use
0X001C	PV1 voltage reference	UINT16	R	V	0.1	Test use

0X001D	PV2voltage reference	UINT16	R	V	0.1	Test use
0X001E	On grid current reference	UINT16	R	A	0.01	Test use
0X001F	Rectified Current Reference	UINT16	R	A	0.01	Test use
0X0020	Load Power	UINT16	R	W	1	
0X0021	Inverter current sampling zero drift	UINT16	R			Test use
0X0022	Inverter voltage sampling zero drift	UINT16	R			Test use
0X0023						
0X0024	Input Voltage Sampling Zero Drift	UINT16	R			Test use
0X0025	Load current sampling zero drift	UINT16	R			Test use
0X0026	Input current sampling zero drift	UINT16	R			Test use
0X0027	Bus current sampling zero drift	UINT16	R			Test use
0X0028	On grid flag bit	UINT16	R			
0X0029						
0X002A						
0X002B						
0X002C						
0X002D						
0X002E						
0X002F						
0X0030	Lithium battery voltage	UINT16	R	V	0.01	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol 5. PYL-A lithium battery protocol Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid.
0X0031	Lithium battery charging and discharging current	INT16	R	A	0.01	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol 5. PYL-A lithium battery protocol

						Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid.
0X0032	Lithium battery temperature	UINT16	R	°C	1	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol 5. PYL-A lithium battery protocol Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid.
0X0033	Lithium battery Capacity SOC474	UINT16	R		1	
0X0034	Lithium battery remaining capacity	UINT16	R	AH	0.01	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol 5. PYL-A lithium battery protocol Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid.
0X0035	Lithium battery rated capacity	UINT16	R	AH	0.01	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol 5. PYL-A lithium battery protocol Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid. .
0X0036	Lithium battery status indicator	UINT16	R		1	According to the lithium battery protocol in the 0X0088 register, select one of the following options: 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: PYL energy lithium battery protocol 4: Pace lithium battery protocol

						5. PYL-A lithium battery protocol Only when a lithium battery is selected will data be returned. When selecting a lead-acid battery, the returned data will be invalid. batteries are selected.
0X0037	PV1 Real-time power	UINT16	R	W	1	
0X0038	PV2 Real-time power 470	UINT16	R	W	1	
0X0039	PV1 Total power generation 468	UINT16	R	KWH	1	
0X003A	PV2 Total power generation	UINT16	R	KWH	1	
0X003B	Real-time power of on grid rectification	UINT16	R	W	1	
0X003C	On grid electricity	UINT16	R	KWH	1	
0X003D	Rectified electric quantity 464	UINT16	R	KWH	1	
0X003E	Battery charging and discharging real-time power	UINT16	R	W	1	
0X003F	Battery charging capacity	UINT16	R	KWH	1	
0X0040	Battery discharge capacity	UINT16	R	KWH	1	
0X0041	Load Power capacity	UINT16	R	KWH	1	
0X0042	Real-time power input from the mains 459	UINT16	R	W	1	
0X0043	Mains electricity input power	UINT16	R	KWH	1	
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.						
.						
0X0060	Device working mode + ID address	UINT16	RW		1	Device ID is low byte: range 1-247 Working mode is high byte; 1:Mains Priority Mode 2:PV Priority Mode 3:Battery Priority Mode
0X0061	Device power	UINT16	RW	W	1	
0X0062	Output Voltage + Frequency	UINT16	RW	V/Hz	1	Output frequency is low byte: 50/60 selectable Output voltage is the high byte: 220/230/240 selectable
0X0063	Charging Priority + Battery Connection	UINT16	RW		1	Whether to connect the battery low byte: 0: Work with battery connected 1: Work without battery connected Charging priority high byte: 0: PV and mains electricity charging

						simultaneously 1: Only pv charging 2: PV charging priority
0X0064	Max charging current + max AC charging current	UINT16	RW	A	1	Low byte of mains charging current:range 2-80A Max charging current high byte:range 2-120A
0X0065	Max rectified current + max on grid current	UINT16	RW	A	1	Max on grid Current Low Byte:range 1-35A Max rectified current high byte:range 1-30A
0X0066	The mains input voltage range + battery nominal voltage	UINT16	RW		1	Battery Nominal Value Low Byte:12V/24V/48V Enter the high byte of the mains power range: 0:165V-280V 1:120V-280V
0X0067	Equalizing voltage	UINT16	RW	V	0.1	12V/24V/48V Setting Range:120V-155V/240V-310V/480V-620V
0X0068	Floating Charge Voltage	UINT16	RW	V	0.1	12V/24V/48V Setting Range:120V-155V/240V-310V/480V-620V
0X0069	Battery high-voltage protection	UINT16	RW	V	0.1	12V/24V/48V Setting Range:Add base on the average charging voltage 28V/56V/112V
0X006A	Battery high voltage alarm	UINT16	RW	V	0.1	12V/24V/48V Setting Range:Add base on the average charging voltage13V/26V/52V
0X006B	Battery high voltage recovery	UINT16	RW	V	0.1	12V/24V/48V setting range:Add base on the average charging voltage:3V/6V/12V
0X006C	Battery low voltage protection	UINT16	RW	V	0.1	12V/24V/48V Setting range:70V-130V/140V-260V/280V-520V
0X006D	Battery low voltage alarm	UINT16	RW	V	0.1	12V/24V/48V Setting range:75V-135V/150V-300V/300V-540V
0X006E	Battery low voltage recovery	UINT16	RW	V	0.1	12V/24V/48V Setting range:75V-150V/150V-300V/300V-600V
0X006F	Overload restart enable + battery low voltage recovery enable	UINT16	RW		1	Battery low voltage recovery enable low byte: 0: Cannot be recovered 1: Can be recovered Load overload recovery enable; 0: Cannot be recovered 1: Can be recovered
0X0070	Inverter priority switch to battery power supply battery voltage	UINT16	RW	V	0.1	12V/24V/48V Setting range:90V-155V/180V-310V/360V-620V
0X0071	Inverter priority switch to mains power supply battery voltage	UINT16	RW	V	0.1	12V/24V/48V Setting range:70V-145V/140V-290V/280V-580V
0X0072	Over-temperature restart + on grid selection	UINT16	RW		1	Low byte of on grid selection: 0: AC power complementary mode 1: On-grid mode 2: Off-grid mode 3: On grid anti-reverse flow mode Overheating restart high byte;

						0: No restart 1: Restartable
0X0073	Reverse-flow protection electric meter current	INT16	R	A	0.01	Select Accessories
0X0074	DC component calibration	INT16	RW	V	1	Range from 100 to-100
0X0075	Neutral grounding and HFP on grid current settings	UINT16	RW		1	Neutral Grounding Setting Low Byte: 0: Neutral not grounded 1: Neutral grounded The high byte is on grid current setting for HFP models: range 0-100
0X0076						
0X0077						
0X0078						
0X0079	Inverter current coefficient	UINT16	RW		1	Test use
0X007A	Inverter voltage coefficient	UINT16	RW		1	Test use
0X007B	DC component coefficient	UINT16	RW		1	Test use
0X007C	bus voltage coefficient	UINT16	RW		1	Test use
0X007D	Input Voltage Coefficient	UINT16	RW		1	Test use
0X007E	Load current coefficient	UINT16	RW		1	Test use
0X007F	Input Current Coefficient	UINT16	RW		1	Test use
0X0080	Battery voltage coefficient	UINT16	RW		1	Test use
0X0081	bus current coefficient	UINT16	RW		1	Test use
0X0082	PV1 current coefficient	UINT16	RW		1	Test use
0X0083	PV1 Voltage coefficient	UINT16	RW		1	Test use
0X0084	PV2 current coefficient	UINT16	RW		1	Test use
0X0085	PV2 Voltage coefficient	UINT16	RW		1	Test use
0X0086	Charging conversion rate + discharging conversion rate	UINT16	RW		1	Low discharge conversion rate byte High charge conversion rate byte
0X0087	BMS Lithium battery address + remote power on/off settings	UINT16	RW		1	Remote power on/off settings low byte: 0: Power off 1: Power on BMS lithium battery address high byte: 0-247
0X0088	Lithium battery shutdown SOC + lithium battery protocol selection	UINT16	RW	%(Lithium battery off SOC)	1	BMS lithium battery protocol low byte: 0: Lead-acid battery 1: Growatt lithium battery protocol 2: Voltronic lithium battery protocol 3: Plyn lithium battery protocol 4: Pace lithium battery protocol 5. Pace-A lithium battery protocol Lithium battery shutdown SOC high byte: range 1-80%

0X0089	Lithium battery charging cut off SOC + lithium battery low SOC recovery	UINT16	RW	%	1	Low SOC byte for lithium battery low voltage recovery: 1-90% High SOC byte for lithium battery charging cut off: 1-100%
0X008A	Inverter priority switch to mains power supply SOC + inverter priority switch to battery power supply SOC	UINT16	RW	%	1	Inverter priority switch to inverter power supply SOC low byte: 1-100% Inverter priority switch to mains power supply SOC high byte: 1-80%
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0X00C0						

System Flag (0X0017) Bit Definition

Position No.	Signal name	Value
Bit0	Power On/Off 449	0: Off, 1: On
Bit1	Buffer Start	0: Not Completed, 1: Buffer Start Completed
Bit2	Mains to Inverter, Rectifier, and Grid-Connected Phase-Locked	0: Abnormal, 1: Normal
Bit3		
Bit4	Bus Soft Start	0: Normal, 1: Abnormal
Bit5	Shutdown/Restart Disabled	0: Normal, 1: Shutdown/Restart Disabled
Bit6	Shutdown/Restart Enabled	0: Normal, 1: Shutdown/Restart Enabled
Bit7	Output Short Circuit	0: Normal, 1: Output Short Circuit
Bit8	Input Voltage Below 50VAC	0: Normal, 1: Below 50VAC
Bit9	Input Frequency	0: Normal, 1: Abnormal
Bit10	Input Voltage Range Selection	0: 165VAC-280VAC, 1: 120VAC-280VAC
Bit11	Sampling Base	0: Normal, 1: Abnormal
Bit12	Anti-Backflow Meter Current Feedback Calculation	0: Not calculated, 1: Calculated
Bit13	Battery Low Voltage Restart	0: Not restarted, 1: Restarted
Bit14	Load Over current Restart	0: Not restarted, 1: Restarted
Bit15	Over temperature Restart	0: Not restarted, 1: Restarted

Inverter Flag (0x0018) Bit Definition

Position No.	Signal name	Value
Bit0	Inverter Soft Start 439	0: Completed, 1: Not completed
Bit1	Mains Input High Voltage	0: Normal, 1: Over voltage
Bit2	Mains Input Low Voltage	0: Normal, 1: Under voltage
Bit3	Busbar High Voltage Restart	0: Restart, 1: Not restarted
Bit4	Inverter to Mains Phase Lock	0: Abnormal, 1: Normal
Bit5	Mains Input	0: Normal, 1: Abnormal
Bit6	Inverter to Mains Delay Switchover	0: Switchover completed, 1: Waiting for switchover
Bit7	Rectifier Soft Start	0: Completed, 1: Soft start not completed
Bit8	Grid-connected soft start	0: Completed, 1: Not completed
Bit9	Inverter temperature	0: Normal, 1: Abnormal
Bit10	Inverter voltage	0: Normal, 1: Abnormal
Bit11	Output load protection	0: Normal, 1: Protection
Bit12	Output load alarm	0: Normal, 1: Alarm
Bit13	Mains-to-inverter delay	0: Normal, 1: Delayed switchover
Bit14	Inverter priority Mains to inverter	0: No switchover, 1: Switchover
Bit15	Off-grid inverter current limit 430	0: Normal, 1: Current limit

DC flag (0X0019) bit definition

Position No.	Signal name	Value
Bit0	Charging soft start 429	0: soft start, 1: soft start completed
Bit1	Battery high voltage	0: Normal, 1: High Voltage
Bit2	Battery high voltage warning	0: Normal, 1: Alarm
Bit3	Battery low voltage protection	0: Normal, 1: Protection
Bit4	Battery low voltage warning	0: Normal, 1: Alarm
Bit5	BUS high voltage protection	0: Normal, 1: Protection
Bit6	BUS low voltage protection	0: Normal, 1: Protection
Bit7	Battery voltage mark 429	0: Normal, 1: Abnormal
Bit8	Battery connection 428	0: Connect to battery, 1: Do not connect to battery
Bit9	BUS voltage 427	0: Normal, 1: Abnormal
Bit10	DC Temperature	0: Normal, 1: Abnormal
Bit11	Mains power priority battery fully charged grid connection 427	0: Feeding without grid connection, 1: Fully charged and grid connected
Bit12	Inverter priority mains battery conversion 426	0: Switch to battery power supply, 1: Switch to mains power supply
Bit13		
Bit14		
Bit15		

MPPT flag (0X001A) bit definition

Position No.	Signal name	Value
Bit0	PV2 high voltage	0: Normal, 1: High voltage
Bit1	PV2 low voltage	0: Normal, 1: low voltage
Bit2	PV1 high voltage	0: Normal, 1: High voltage
Bit3	PV1 low voltage	0: Normal, 1: low voltage
Bit4	MPPT2 soft start	0: Not completed, 1: Completed
Bit5	MPPT1 soft start	0: Not completed, 1: Completed
Bit6	PV1 temperature abnormality	0: Normal, 1: Abnormal
Bit7	MPPT1 normal working 414	0: Abnormal, 1: Normal
Bit8	PV1 voltage 412	0: Abnormal, 1: Normal
Bit9	PV2 voltage 412	0: Abnormal, 1: Normal
Bit10	MPPT2 normal working 414	0: Abnormal, 1: Normal
Bit11		
Bit12		
Bit13		
Bit14		
Bit15		

On-Grid connection flag (0X0028) bit definition

Bit No.	Signal name	Value
Bit0	On-Grid voltage low 409	0: Normal, 1: Low Voltage
Bit1	On-Grid voltage high	0: Normal, 1: Over voltage
Bit2	On-Grid voltage	0: Normal, 1: Abnormal
Bit3	On-Grid frequency	0: Normal, 1: Abnormal
Bit4	PV priority mode on-grid operation 405	0: Can't on-grid; 1: Can on- grid
Bit5		
Bit6		
Bit7		
Bit8		
Bit9		
Bit10		
Bit11		
Bit12		
Bit13		
Bit14		
Bit15		

Note: Lithium battery status alarm information bit0-bit10, other bits are ignored, the lithium battery alarm information is displayed only when the lithium battery protocol option is not 0, and the lithium battery alarm information is not displayed when it is 0

Warning Code				
Address	Content(Binary)	State	Description	Recovery Mechanism
0X0022	Bit 0	0	Normal	Recover after discharging or when the voltage is lower than the single cell over voltage alarm value (lithium iron /ternary battery)
		1	Single cell over voltage alarm	
	Bit 1	0	Normal	Recover after charging or when the voltage exceeds the single cell under voltage alarm value (lithium iron /ternary battery)
		1	Single cell under voltage alarm	
	Bit 2	0	Normal	Discharge or voltage is lower than the total voltage over voltage alarm value recovery (lithium iron /ternary battery)
		1	Total voltage over voltage alarm	
	Bit 3	0	Normal	Charging or voltage higher than the total voltage under voltage alarm value recovery (lithium iron /ternary battery)
		1	Total voltage under voltage alarm	
	Bit 4	0	Normal	The current is lower than the discharge over current alarm value
		1	Discharge over current alarm	
	Bit 5	0	Normal	The current is lower than the charging over current warning value
		1	charge over current alarm	
	Bit 6	0	Normal	Temperature is lower than the discharge high temperature warning value (℃)
		1	Discharge high temperature alarm	
	Bit 7	0	Normal	The Temperature is higher than the discharge low temperature warning value (℃)
		1	Discharge low temperature alarm	
	Bit 8	0	Normal	Temperature is lower than the charging high temperature warning value (℃)
		1	Charging high temperature alarm	
	Bit 9	0	Normal	Temperature is higher than the charging low temperature warning value (℃)
		1	Charging low temperature alarm	
	Bit 10	0	Normal	Temperature is lower than MOS high temperature warning value (℃)
		1	MOS high temperature alarm	
		1	Environment low temperature alarm	(℃)
	Bit 13	0	Normal	Total voltage is higher than the system shutdown/lock voltage alarm value (℃)
		1	System low voltage shutdown warning	
0X0036	Bit 14-Bit 15	Battery Type		00: Lithium iron battery 01: Ternary battery 10: Lithium titanate battery 11: Reserve

Display terminal value (0X002A) bit definition

Bit No.	Signal name	Value
Bit0	Data storage EEPROM	0: Normal, 1: Error
Bit1	Communication with main control unit	0: Normal, 1: Error
Bit2	Parameter setting s	0: Normal, 1: Abnormal
Bit3	Remote power on/off s	0: On, 1: Off
Bit4	Communicating with lithium battery	0: Normal, 1: Abnormal
Bit5		
Bit6		
Bit7		
Bit8		
Bit9		
Bit10		
Bit11		
Bit12		
Bit13		
Bit14		
Bit15		